

Construction Safety Detector

Computer vision for detecting hard hats and safety vests
on construction sites

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Midterm Project Proposal



INPUT → DETECT → ALERT

The Problem

A real-world safety check that can be slow and inconsistent when done only by eye.

01

Real-world problem

Workers can enter active construction areas without required PPE, especially hard hats or high-visibility vests.

02

Who cares?

Workers, safety managers, supervisors, contractors, and companies responsible for workplace safety.

03

Why it matters

Missing PPE can increase injury risk and makes safety monitoring harder across busy or large job sites.

04

What will help

A vision system that flags whether key PPE is visible in a photo so people can review problems quickly.

Project goal

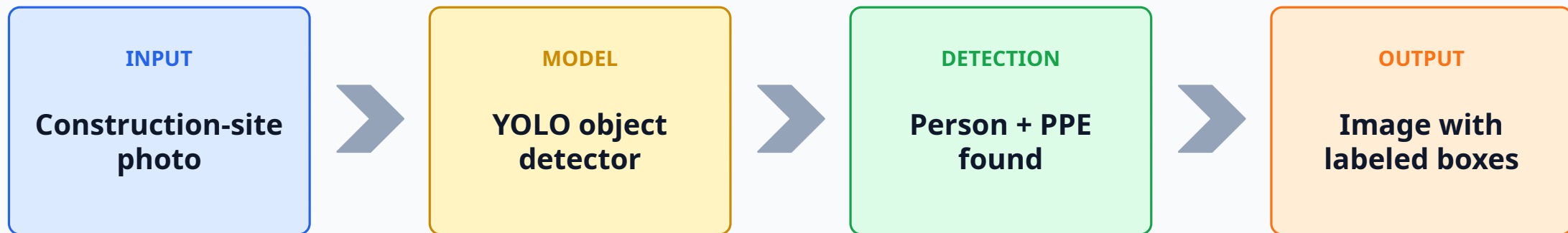
Use computer vision to automatically detect whether workers and required safety equipment are visible in construction-site images, making PPE checks faster and easier to review.

Your Solution

A simple image-to-detection workflow designed for a clear final demonstration.

One-sentence solution

A computer-vision application will analyze a construction-site image, detect people and PPE, and display labeled bounding boxes for safety review.



Final demo: upload a test image → run the detector → review bounding boxes and confidence scores.

Technical Approach

Object detection is the best fit because the system must identify both what an item is and where it appears.

CV CV technique

Object detection

The model will locate multiple items in one image and draw a bounding box around each detected object.

AI Model

YOLO (pretrained)

Transfer learning will let the project start with a model that already understands general visual features.

PY Framework

Ultralytics + Python

Training and inference can run in Google Colab with a notebook-based workflow.

Why this approach?

Classification would label an entire image. This project needs to find several workers and PPE items in the same scene, so bounding-box object detection is more useful.

Data Plan

Public annotated images will be an ingredient for training—not the end product.

Planned source

Roboflow Universe Construction Site Safety dataset

- Public object-detection dataset
- Roughly 717 annotated images
- Includes construction scenes and PPE labels
- Project will focus on the classes needed for this application

Source checked Sept. 2026 • CC BY 4.0

Target labels

Person

Hardhat

Safety Vest

NO-Hardhat

NO-Safety Vest

Preparation

1. Inspect labels
2. Remove unusable images
3. Keep target classes
4. Split train / validation / test
5. Resize to model input

Success Metrics

Two measurable targets will define whether the proof of concept is successful.

PRIMARY METRIC

mAP@50

$\geq 75\%$

Target detection quality on the validation/test images.

SECONDARY METRIC

Inference time

$< 1 \text{ sec} / \text{image}$

Fast enough for a smooth single-image classroom demo.

Success = useful detections + a demo that responds quickly enough to review results in real time.

Milestone Plan

Six phases move the project from proposal to a working final demonstration.

1	Week 1	Finalize scope	Confirm dataset, labels, repo structure
2	Week 2	Prepare data	Inspect annotations and create train/val/test split
3	Week 3	Train baseline	Fine-tune pretrained YOLO model
4	Week 4	Evaluate + improve	Review errors, tune settings, retrain if needed
5	Week 5	Build demo	Create simple image upload and detection workflow
6	Week 6	Final QA	Test, document, record demo, submit project

Timeline can be adjusted to match the remaining course weeks.

Risks and Resources

The project stays practical by planning for two likely problems before training begins.



Risk 1 • Labels or images are inconsistent

Plan B: limit the training set to the five target classes, remove unusable examples, and supplement with another compatible public PPE dataset if needed.



Risk 2 • Training is too slow or unstable

Plan B: use a smaller pretrained YOLO model, reduce image size/epochs, and focus on a reliable proof of concept instead of a larger model.

Resources

Compute	Google Colab (GPU when available)
Software	Python • Ultralytics YOLO • Jupyter/Colab
Code + docs	GitHub repository
Estimated cost	\$0 using free/open-source resources

Final proof of concept

A user submits a construction image and receives labeled detections showing people, hard hats, safety vests, and missing-PPE classes.